

Rachel Abraham

rachel6ad@gmail.com | 647-381-4494 | Ann Arbor, MI | www.linkedin.com/in/rachel-abraham

EDUCATION

University of Michigan - Ann Arbor

Master of Applied Data Science (GPA 4.0)

Ann Arbor, MI

May 2024

University of Waterloo

Bachelor of Science in Organic Chemistry (GPA 3.2)

Waterloo, ON

June 2017

EXPERIENCE

Conversational AI Consultant

Telus Digital

June 2024 - Present

Part-time

- Designed and implemented AI-powered virtual assistant experiences using Dialogflow CX and Vertex AI, increasing customer engagement by 30%.
- Optimized guided conversational flows and prompt engineering strategies, reducing query response times by 25%.
- Leveraged Google Cloud services (BigQuery, Cloud Functions) for performance analytics and continuous improvement of chatbot performance.

Machine Learning Researcher

University of Michigan

September 2022 - Present

Part-time

- Developed and fine-tuned large language models for conversational applications, achieving a 20% increase in dialogue accuracy.
- Engineered data pipelines to preprocess multi-modal data (text, audio, video), ensuring high-quality inputs for NLP and chatbot models.
- Integrated advanced embedding models and retrieval-augmented generation (RAG) techniques to support dynamic conversational responses.

AI & Prompt Engineering Instructor

BusyQA

November 2023 - March 2024

Seasonal

- Delivered training on conversational AI design, including guided flow creation and prompt engineering best practices, achieving a 100% student satisfaction rate.
- Developed hands-on labs for chatbot development using Dialogflow CX, fostering practical skills in conversational AI integration.

Data Scientist

Triton Canada

September 2022 - September 2023

Full-time

- Implemented ETL pipelines and KPI dashboards with SQL and Looker Studio to analyze chatbot user interactions, boosting operational efficiency by 40%.
- Collaborated with cloud and engineering teams to integrate scalable AI solutions using Google Cloud technologies.

TECHNICAL PROJECTS

Conversational AI Virtual Assistant | Dialogflow CX, Vertex AI, Python

March 2025

- Designed and developed an end-to-end conversational agent that handles customer queries, increasing engagement by 30%.
- Implemented guided conversational flows and prompt engineering strategies to optimize dialogue performance.

Chatbot Performance Analytics | BigQuery, Looker Studio, Python

January 2025

- Developed analytical dashboards to monitor and evaluate chatbot performance, reducing average response times by 20%.
- Integrated Google Cloud data services to streamline real-time data collection and performance analysis.

RAG-Enhanced Conversational Model | LLM Fine-Tuning, RAG, Terraform

December 2024

- Built a retrieval-augmented generation framework to support dynamic conversational responses, improving model accuracy by 15%.
- Deployed the model using infrastructure as code (Terraform), ensuring reproducible and scalable cloud deployment.

TECHNICAL SKILLS

Programming Languages: Python, JavaScript, SQL

Conversational AI: Dialogflow CX, Vertex AI, Guided Conversational Flows, Prompt Engineering

AI Techniques: Generative AI, LLM Fine-Tuning, NLP, Retrieval-Augmented Generation (RAG)

Tools & Cloud Services: Google Cloud (BigQuery, Cloud Functions), Terraform, Looker Studio, Docker, AWS

LEADERSHIP & AWARDS

Scholarship: University of Michigan, 2023

Awards: Distinguished Teaching Assistant Award, University of Waterloo, 2016

Conference: Grace Hopper Conference, 2023